Qinghua Zhou

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Contact Information

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Intro

About me: I am a researcher and engineer working on the safety and alignment of artificial intelligence systems. I have extensive experience in dynamic collaborations with mathematicians, archaeologists and clinicians, bridging frontier AI mathematical theory to practical AI applications, providing deterministic methods to affect model behavior (e.g., de-bias, stain, lock, edit or attack) with minimal optimization and negligible cost.

Academics

King's College London, Department of Mathematics — 2022–2025
Postdoctoral Research Associate in Methods and Algorithms of AI.

University of Leicester, School of Computing and Mathematical Sciences, — 2022
Postdoctoral Research Associate with Ivan Tyukin.

University of Leicester, School of Computing and Mathematical Sciences, — 2019–2023
PhD in Computer Science supervised by Yudong Zhang and Ivan Tyukin.

University of Sydney School of Mathematics and Statistics, and School of Physics, — 2016–2018
BSc in Science with majors in Physics and Applied Mathematics

Experience

Industry Collaboration with TG-0 London — 2024
Development of reservoir computing algorithms for specific hardware applications.

Graduate Teaching Assistant with School of Computing and Mathematical Sciences — 2019–2021
AFHEA accredited. Delivery of a range of undergraduate and master's courses.

Research Assistant Coordinator with Department of Cardiovascular Sciences, UoL — 2019–2020
Led a group of research assistants in ML-based knowledge discovery of gene expression data.

Select Publications

Harnessing non-adversarial robustness in large language models
ICML, Seoul 2026 | Spotlight Paper (Top 2.2%)

LLMs can falter when input prompts have slight text or format differences. This is due to a shift in the model's internal signal — like a scale that's slightly off-balance. The "scale" can be re-adjusted with a small correction.

Through analysis, we trace fragility from semantically-neutral perturbations to systematic biases in module outputs. A cheap closed-form debiasing can be applied with no retraining and often no labeled data.

Stealth edits for provably fixing or attacking large language models
NeurIPS, Vancouver 2024 | Huggingface Live Demo | SIAM Front Page

LLMs sometimes make factual errors or bad responses, which you can surgically patch by tweaking a few neurons' weights, with guarantees that other answers stay untouched. Unfortunately, this also means an attacker can plant a hidden trigger that's nearly impossible to detect.

Through in-depth understanding of LLM and its latent spaces, we show that a single quantity, the separability-based intrinsic dimensionality of a model's latent features, provably governs the selectivity of model editing.

Staining and locking computer vision models without retraining
ICCV, Hawai'i 2025 | Streamlit Live Demo | UK Patent

To protect your trained model, there are ways to "stain" one with a hidden fingerprint that proves ownership, or "lock" one so that a thief who copies it gets near-useless performance. These protections can be installed directly into the model itself without retraining.

We add highly selective detector neurons and disruptor mechanisms directly into the weights and structures of a model. Exploiting feature-space concentration, we gain computable worst-case false-positive bounds.

Technical Skills

Languages: Python, Matlab (Expert). Mathematica (Proficient), Rust, Typescript, SQL (Comfortable).

ML & DL Frameworks: Pytorch, Tensorflow, NumPy, Pandas, Scikit-Learn.

CV & NLP Stacks: OpenCV, PIL, Nibabel, FSL, vLLM, LangChain, RAGs.

Formal Verification: Lean (Comfortable), Coq, ISAbelle (Beginner).

Other Interesting Works

Neuromorphic tuning of feature spaces to overcome the challenge of low-sample high-dimensional data

This work considers the scenario of mesoscopic sample-size, and based on theoretical foundations of few-shot learning, presents a neuromorphic algorithm to improve latent spaces of pre-trained models.

Quasi-orthogonality and intrinsic dimensions as principled measures of learning and generalisation

This work examines the correlations between the accuracies of trained networks and their initial states and provides a new perspective for zero-shot NAS for finding better architectures.

How adversarial attacks can disrupt seemingly stable accurate classifiers

This work demonstrates that for classifiers on high-dimensional data we have simultaneous susceptibility to adversarial attacks and robustness to random perturbations.

Multiple-instance ensemble for construction of deep heterogeneous committees for high-dimensional low-sample-size data

This work introduces a novel class of stacking methods that utilize attention pooling mechanisms for the construction of both ensembles and cascades for low-sample size medical imaging datasets.

More publications can be found in the list below, or on my Google Scholar Page

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Grants & Accred.

GTA PhD Scholarship, University of Leicester

Lambda Research Grant 2025

Associate Fellow of the Higher Education Academy (AFHEA)

Citations.

Total 1122, **H-index** 11

List of Publications

Zhou, Q., Aleshina,E., Lovyagin,A., Somov, O., Seleznyov,M., Panchenko, A., Oseledets, I., Tutubalina,E., Tyukin, I.Y.. 2026. Harnessing non-adversarial robustness in large language models. In proceedings of *International Conference of Machine Learning (ICML)*.

Sutton, O.J.*, **Zhou, Q.***, Leete, F., Gorban, A., Tyukin, I.Y. 2025. Staining and locking computer vision models without retraining. In proceedings of *International Conference on Computer Vision (ICCV)*.

Sutton, O.J.*, **Zhou, Q.***, Wang, W., Higham , D.J., Gorban, A.N., Bastounis, A., Tyukin, I.Y. 2024. Stealth edits for provably fixing or attacking large language models. In proceedings of *Conference on Neural Information Processing Systems (NeurIPS)*.

Sutton, O.J., **Zhou, Q.**, Tyukin, I.Y., Gorban, A.N., Bastounis, A. and Higham, D.J., 2024. How adversarial attacks can disrupt seemingly stable accurate classifiers. *Neural Networks*, 180, 106711.

Tyukin, I.Y., Tykina, T., van Helden, D.P., Zheng, Ze., Mirkes, E.M., Sutton, O., **Zhou, Q.**, Gorban, A.N., Allison, P., 2024. Coping with AI errors with provable guarantees *Information Sciences*, 678, 106711.

Tyukin, I.Y., Tykina, T., van Helden, D.P., Zheng, Ze., Mirkes, E.M., Sutton, O., **Zhou, Q.**, Gorban, A.N., Allison, P. 2024. Weakly Supervised Learners for Correction of AI Errors with Provable Performance Guarantees. In proceedings of *International Joint Conference on Neural Networks*.

Zhu, H., Wang, W., Ulidowski, I., **Zhou, Q.**, Wang, S., Chen, H., Zhang, Y., 2023. MEEDNets: Medical Image Classification via Ensemble Bio-inspired Evolutionary DenseNets. *Knowledge-Based Systems*, 280, 111035.

Zhou, Q., Wang, S., Zhu, H., Zhang, X., & Zhang, Y. (2023). Multiple-instance ensemble for construction of deep heterogeneous committees for high-dimensional low-sample-size data. *Neural Networks*, 167, 380-399.

Sutton, O., **Zhou, Q.**, Gorban, A.N., Tyukin, I.Y., 2023. Relative intrinsic dimensionality is intrinsic to learning. In proceedings of *International Conference on Artificial Neural Networks (ICANN)*.

Bastounis, A., Gorban, A.N., Hansen, A.C., Higham, D.J., Prokhorov, D., Sutton, O., Tyukin, I.Y., **Zhou, Q.**, 2023. The Boundaries of Verifiable Accuracy, Robustness, and Generalisation in Deep Learning. In proceedings of *International Conference on Artificial Neural Networks (ICANN)*.

Zhou, Q., Sutton, O., Zhang, Y.D., Makarov, V.A., Gorban, A.N., Tyukin, I.Y., 2023. Neuromorphic tuning of feature spaces to overcome the challenge of low-sample high-dimensional data. In proceedings with oral presentation in *International Joint Conference on Neural Networks*.

Zhou, Q., Gorban, A.N., Mirkes, E.M., Bac, J., Zinovyev, A., Tyukin, I.Y., 2022. Quasi-orthogonality and intrinsic dimensions as measures of learning and generalisation. In proceedings with oral presentation in *International Joint Conference on Neural Networks*.

*Equal contributions, alphabetical order

Zhou, Q., Wang, S., Zhang, X., Zhang Y.D., 2021. WVALE: Weak Variational Autoencoder for Localisation and Enhancement of COVID-19 Lung Infections. *Computer Methods and Programs in Biomedicine*, 221, 106883. + Physicsworld Front Page Article.

Wang, S., Stapathy, S.C., **Zhou, Q.**, Zhang, X., Zhang, Y., 2022. Secondary pulmonary tuberculosis identification via pseudo-Zernike moment and deep stacked sparse autoencoders. *Journal of Grid Computing*, 20, 1-16.

Yu, X., **Zhou, Q.**, Wang, S. and Zhang, Y.D., 2021. A systematic survey of deep learning in breast cancer. *International Journal of Intelligent Systems*, 37(1), 152-216.

Zhou, Q., Zhu, H., Zhang, X. and Zhang, Y., 2021, December. Knowledge distillation for secondary pulmonary tuberculosis classification ensemble. In Proceedings of the *14th IEEE/ACM International Conference on Utility and Cloud Computing Companion*.

Wang, S., Celebi, M.E., Zhang, Y.D., Yu, X., Lu, S., Yao, X., **Zhou, Q.**, Miguel, M.G., Tian, Y., Gorriz, J.M. and Tyukin, I., 2021. Advances in Data Preprocessing for Biomedical Data Fusion: An Overview of the Methods, Challenges, and Prospects. *Information Fusion*, 76, 376-421.

Zhou, Q., Zhang, X. and Zhang, Y.D., 2021. Ensemble learning with attention-based multiple instance pooling for classification of SPT. *IEEE Transactions on Circuits and Systems II*, 69(3), 1927-1931.

Tyukin, I.T., Gorban, A.N., Alkhudaydi, M.H., and **Zhou, Q.**, 2021. Demystification of Few-shot and One-shot Learning. In Proceedings of the *International Joint Conference on Neural Networks*.

Wang, S.H., Wu, K., Chu, T., Fernandes, S.L., **Zhou, Q.**, Zhang, Y.D. and Sun, J., 2021. SOSPCNN: Structurally Optimized Stochastic Pooling Convolutional Neural Network for Tetralogy of Fallot Recognition. *Wireless Communications and Mobile Computing*, 2021(1), 5792975.

Chen, X., **Zhou, Q.**, Lan, R., Wang, S.H., Zhang, Y.D. and Luo, X., 2021. Sensorineural hearing loss classification via deep-HLNet and few-shot learning. *Multimedia Tools and Applications*, 80(2), 2109-2122.

Zhu, M., Zhao, D., Wang, Y., **Zhou, Q.**, Wang, S., Mo, X., Yang, M. and Sun, Y., 2020. Multi-Slice Radiomic Analysis of Apparent Diffusion Coefficient Metrics Improves Evaluation of Brain Alterations in Neonates With Congenital Heart Diseases. *Frontiers in neurology*, 11, 586518.

Zhang, Y.D., Dong, Z., Wang, S.H., Yu, X., Yao, X., **Zhou, Q.**, Hu, H., Li, M., Jiménez-Mesa, C., Ramirez, J. and Martinez, F.J., 2020. Advances in multimodal data fusion in neuroimaging: Overview, challenges, and novel orientation. *Information Fusion*, 64, 149-187.

Kang, C., Li, Y., Novak, D., Zhang, Y., **Zhou, Q.** and Hu, Y., 2020. Brain Networks of Maintenance, Inhibition and Disinhibition During Working Memory. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 28(7), 1518-1527.

Zhou, Q., Mirkes, EM., Zhang, Y., Tyukin, I., Gorban, AN., Yu, X., Wang S., Charlton M., Coats, T., Sims MR., Thompson, JP. 2020, Machine learning technique for severity classification in sepsis patients. *Intensive Care Medicine Experimental 2020*, 8(Suppl 2):73 ID 000473.

Zhou, Q., Zhu, H., Yao, X., Zhang, Y., Murphy, G., Adebayo, A., Wozniak, M., 2020. Obesity-driven differences in gene expression – a machine learning approach. Abstract at *SCTS 2020 conference*.

References — Available upon request.